Lecture 10

L'Hôpital and the shape of a graph

Uses.

Summary 5: One-sided and infinite limits have explicit input tests. Two available side limits give a two-sided limit exactly when they agree.
Summary 6: The exponential tends to infinity at $+\infty$. Positive-base powers may be treated through their logarithms.
Summary 7: A continuous function on a closed bounded interval attains its extreme values.
Summary 8: The derivative rules and elementary table hold on their stated domains.
Summary 9: Fermat, Rolle and Lagrange require their interior and endpoint hypotheses. Derivative signs give monotonicity on intervals.

Objectives. *Cauchy's theorem compares changes in two functions. It proves a rule for zero-over-zero limits. Derivative signs then locate extrema and describe convexity. We combine these tests with domain and asymptote calculations in one graph investigation, and use them to choose the dimensions of a box.*

By the end of this lecture you will be able to:

- Prove Cauchy's theorem and the one-sided zero-over-zero form of L'Hôpital's rule.
- Apply L'Hôpital's rule only after checking the limiting form and denominator derivative.
- Prove and use derivative tests for extrema and convexity.
- Investigate a rational graph and solve an optimisation problem on its physical domain.

10.1 Cauchy's mean value theorem

Theorem 10.1 (Cauchy's mean value theorem)**.** *Let $a < b$. Suppose f and g are continuous on $[a, b]$ and differentiable on (a, b). Then some $c \in (a, b)$ satisfies*

$$[f(b) - f(a)]g'(c) = [g(b) - g(a)]f'(c).$$

If g' is nonzero everywhere on (a, b), then $g(b) \neq g(a)$ and

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}.$$

Proof. Define $H(x) = [f(b) - f(a)]g(x) - [g(b) - g(a)]f(x)$. It is continuous on the closed interval and differentiable inside. Expanding its endpoint difference gives $H(b) - H(a) = 0$.

Rolle's theorem supplies $c \in (a,b)$ with $H'(c) = 0$. The derivative rules give the first displayed identity. If g' never vanishes, Rolle's theorem also rules out $g(a) = g(b)$. Both divisors in the second identity are then nonzero, so division is valid. ∎

Taking $g(x) = x$ recovers Lagrange's theorem, since $g'(c) = 1$ and $g(b) - g(a) = b - a$. Keeping a second function instead allows us to compare two changes which both tend to zero.

10.2 L'Hôpital's rule and its hypotheses

Theorem 10.2 (One-sided zero-over-zero rule)**.** *Let $a < b$ be real numbers and let f and g be differentiable on (a,b). Suppose $f(x) \to 0$ and $g(x) \to 0$ as $x \to a+$, and $g'(x) \neq 0$ on (a,b). If $f'(x)/g'(x) \to L$ as $x \to a+$, where L is real or one of the symbols $+\infty$ and $-\infty$, then $g(x) \neq 0$ on (a,b) and*

$$\frac{f(x)}{g(x)} \longrightarrow L \qquad (x \to a+).$$

Proof. Assign $f(a) = g(a) = 0$. Their assumed limits make these extensions continuous at a. For any $x \in (a,b)$, if $g(x) = 0$, Rolle's theorem on $[a,x]$ would give an interior zero of g'. This is excluded, so $g(x) \neq 0$. Cauchy's theorem on $[a,x]$ now gives a point $c_x \in (a,x)$ with

$$\frac{f(x)}{g(x)} = \frac{f'(c_x)}{g'(c_x)}.$$

For any requested tolerance, choose a right-neighbourhood of a where the derivative ratio is within that tolerance of a finite L. If x is in this neighbourhood, then $a < c_x < x$ puts c_x there too. The identity gives the same bound for $f(x)/g(x)$. If L is infinite, choose the derivative-ratio neighbourhood for a requested output bound instead. The same identity proves that bound for the original ratio. These are exactly the required limit tests. ∎

The left-hand version follows by reversing the input with $x = -t$. The two derivative factors -1 cancel in their ratio. If both sides meet the hypotheses with a common L, the one-sided criterion gives the two-sided conclusion. A zero-over-zero limit at $+\infty$ or $-\infty$ reduces to a finite one-sided limit by $x = 1/t$ or $x = -1/t$ with $t \to 0+$. The common nonzero derivative factor of this substitution again cancels.

For example, as $x \to 0$ with $x > -1$,

$$\frac{\ln(1+x) - x}{x^2} \longrightarrow -\frac{1}{2}.$$

The numerator and denominator tend to zero. On each deleted side, the denominator derivative $2x$ is nonzero. The derivative ratio is

$$\frac{1/(1+x) - 1}{2x} = -\frac{1}{2(1+x)} \longrightarrow -\frac{1}{2}.$$

The proved rule applies on both sides, giving the common value. This calculation does not differentiate the quotient itself. It compares the derivatives of its numerator and denominator separately.

We also use the *infinity-over-infinity form*, cited here from Zorich, Volume I, §5.4. On a one-sided finite or infinite approach interval, assume f and g are differentiable, both tend

to an infinite limit, g' never vanishes, and $f'/g' \to L$ with L finite or infinite. Then $f/g \to L$. This form is *cited*, rather than proved above. The denominator is eventually nonzero because its absolute value tends to infinity.

For example, $\ln x/x \to 0$ as $x \to +\infty$ by the cited form: both functions tend to $+\infty$ and their derivative ratio is $1/x \to 0$. The logarithm's infinite limit follows from its order: $x > e^M$ implies $\ln x > M$. The evaluation uses the cited rule. Likewise, as $x \to 0+$,

$$x \ln x = \frac{\ln x}{1/x} \longrightarrow 0,$$

because the numerator tends to $-\infty$, the denominator to $+\infty$, and their derivative ratio is $(1/x)/(-1/x^2) = -x \to 0$. This evaluation also retains the cited dependency.

Both the limiting form and the derivative-ratio limit matter. For $(1 + x^2)/(1 + x)$ at zero, substitution gives the limit 1. The derivative ratio tends to zero, but the original form is neither zero-over-zero nor infinity-over-infinity. No rule licenses that replacement. Conversely, the ratio $(x + \sin x)/x$ tends to 1 at $+\infty$, since $|\sin x/x| \le 1/x$. Its derivative ratio $1 + \cos x$ has no limit: it equals 2 at $2\pi n$ and zero at $(2n + 1)\pi$. Failure of a derivative-ratio limit does not prove failure of the original limit.

10.3 Tests for local extrema

A *critical number* is an interior domain input where a function's derivative is zero or does not exist. Fermat's theorem puts every interior local extremum among these candidates. Domain endpoints must be checked separately in a global optimisation. A point outside the domain is not a critical number. An input c with $f'(c) = 0$ is called *stationary*.

A *strict* local maximum requires $f(x) < f(c)$ at all nearby domain points other than c. Strict local minima use the reverse inequality.

Proposition 10.3 (First-derivative test)**.** *Suppose f is continuous at an interior input c and differentiable in a deleted neighbourhood. If $f' > 0$ just to the left and $f' < 0$ just to the right, then c is a strict local maximum. Reversing these signs gives a strict local minimum. If both signs are positive, or both negative, there is no local extremum at c.*

Proof. In the maximum case, choose $x < c < y$ close enough for the sign assumptions. The mean value theorem on $[x, c]$ gives $f(c) - f(x) > 0$. On $[c, y]$ it gives $f(y) - f(c) < 0$. Thus all nearby distinct values are below $f(c)$. Apply this result to $-f$ for the minimum case. If both signs are positive, the same reasoning gives $f(x) < f(c) < f(y)$, ruling out either extremum. Negative signs reverse both comparisons and give the same exclusion. ∎

For example, a positive-to-negative derivative change gives more than the equation $f'(c) = 0$: it compares the actual surrounding values. The test can also apply to a corner where $f'(c)$ does not exist.

Proposition 10.4 (Second-derivative test)**.** *Suppose f is differentiable near c, f' is differentiable at c, and $f'(c) = 0$. If $f''(c) > 0$, then c is a strict local minimum. If $f''(c) < 0$, it is a strict local maximum.*

Proof. When $f''(c) > 0$, the definition gives

$$\frac{f'(c + h) - f'(c)}{h} \longrightarrow f''(c) > 0.$$

This quotient is positive for every sufficiently small nonzero h. Since $f'(c) = 0$, we obtain $f'(c+h) < 0$ for negative h and $f'(c+h) > 0$ for positive h. The first-derivative test gives the minimum. For $f''(c) < 0$, apply the proved case to $-f$. ∎

The value $f''(c) = 0$ is inconclusive. At zero, x^4 has a minimum, $-x^4$ has a maximum and x^3 has neither, although their first and second derivatives there are zero.

10.4 Convexity from derivative signs

Definition 10.5 (Convexity and concavity)**.** A real function f on an interval I is *convex* if

$$f((1-t)x + ty) \le (1-t)f(x) + tf(y) \qquad (x, y \in I,\ 0 \le t \le 1).$$

It is *concave* if these inequalities are reversed.

The convex inequality says that the graph lies at or below each endpoint secant. A differentiable convex function has slopes that do not decrease. The following result makes this precise.

Theorem 10.6 (Convexity criterion)**.** *For a differentiable function on an open interval, convexity is equivalent to its derivative being nondecreasing. For a twice differentiable function on that interval, convexity is equivalent to $f'' \ge 0$. Concavity reverses these inequalities.*

Proof. Suppose first that f is convex. For $x < z < y$ in the interval, substituting $z = (1-t)x+ty$ in the defining inequality and rearranging gives

$$\frac{f(z) - f(x)}{z - x} \le \frac{f(y) - f(x)}{y - x} \le \frac{f(y) - f(z)}{y - z}.$$

Letting $z \to x+$ in the first inequality gives $f'(x) \le [f(y) - f(x)]/(y - x)$. Letting $z \to y-$ in the second gives that this same secant slope is at most $f'(y)$. Therefore $f'(x) \le f'(y)$ for every $x < y$.

Conversely, suppose f' is nondecreasing. For $x < z < y$, the mean value theorem gives $u \in (x, z)$ and $v \in (z, y)$ with

$$\frac{f(z) - f(x)}{z - x} = f'(u) \le f'(v) = \frac{f(y) - f(z)}{y - z}.$$

Multiplying by the positive denominators and collecting terms gives

$$f(z) \le \frac{y - z}{y - x} f(x) + \frac{z - x}{y - x} f(y).$$

This is the convex inequality for interior weights. Endpoint weights and equal inputs give equality, and exchanging x and y covers their opposite order. Thus f is convex.

If f is twice differentiable, Lecture 9's derivative-sign theorem and its converse say that f' is nondecreasing exactly when $f'' \ge 0$. This proves the second criterion. Applying both conclusions to $-f$ gives the concavity criteria. ∎

An *inflection point* is a point $(c, f(c))$ with the following properties. For some $\delta > 0$, the function is defined on $(c-\delta, c+\delta)$ and is continuous at c. It is convex on $(c-\delta, c]$ and concave on $[c, c+\delta)$, or conversely. In addition, it is neither convex throughout nor concave throughout any open interval around c contained in that neighbourhood. An affine function $f(x) = mx + b$ is both convex and concave throughout every interval, so it has no inflection points under this definition.

Suppose f is continuous at c and twice differentiable on both adjacent open sides. Opposite strict signs of f'' on those sides give an inflection point. The convexity criterion applies separately on each side, and continuity extends its inequality to the endpoint c. A full convex neighbourhood would force $f'' \geq 0$ on the side where $f'' < 0$, a contradiction. A full concave neighbourhood would similarly contradict the side where $f'' > 0$. Thus the required change holds. The value $f''(c) = 0$ alone does not establish it: x^4 remains convex across zero.

10.5 Asymptotes and one full graph investigation

The line $x = a$ is a *vertical asymptote* if a one-sided limit of f at a is $+\infty$ or $-\infty$. An *affine asymptote* at $+\infty$ or $-\infty$ is a line $y = mx + b$ for which $f(x) - (mx + b) \to 0$ on that approach. It is *horizontal* if $m = 0$ and *oblique* if $m \neq 0$. If it exists, its coefficients satisfy

$$m = \lim \frac{f(x)}{x}, \qquad b = \lim(f(x) - mx).$$

Indeed, dividing $f(x) = mx + b + r(x)$ with $r(x) \to 0$ by x gives the first limit, and subtraction gives the second. Conversely, a finite second limit b gives exactly the required vanishing error for that m. A slope limit alone is insufficient: $\sqrt{x}/x \to 0$ at $+\infty$, but $\sqrt{x}$ has no finite horizontal asymptote.

Figure 10.1 shows the function investigated next.

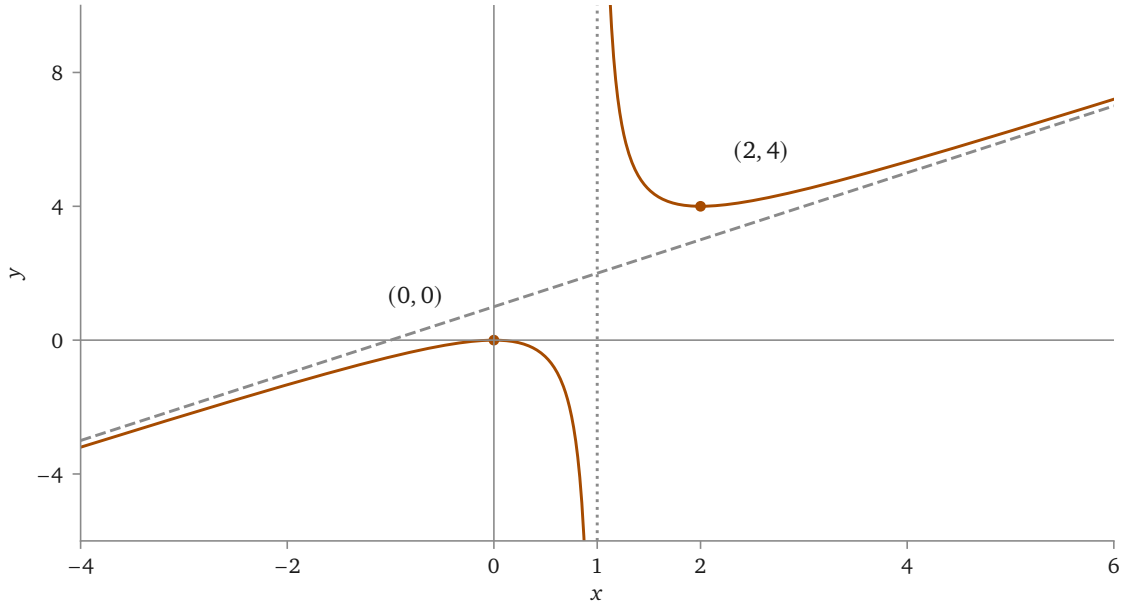


Figure 10.1: The graph of $f(x) = x^2/(x-1)$. The dotted vertical line is $x = 1$. The dashed oblique line is $y = x + 1$. The marked points are the local extrema.

Consider $f(x) = x^2/(x-1)$ on $\mathbb{R} \setminus \{1\}$. It is continuous and differentiable on each of its domain intervals. Long division gives the exact identity

$$f(x) = x + 1 + \frac{1}{x-1}.$$

Thus $y = x + 1$ is an oblique asymptote at both infinities. As $x \to 1-$ the last term tends to $-\infty$, and as $x \to 1+$ it tends to $+\infty$, so $x = 1$ is a vertical asymptote. The only axis intersection is $(0, 0)$.

Differentiating the identity gives

$$f'(x) = 1 - \frac{1}{(x-1)^2} = \frac{x(x-2)}{(x-1)^2}, \qquad f''(x) = \frac{2}{(x-1)^3}.$$

The critical numbers are 0 and 2. The missing input 1 is not critical. The first-derivative signs and monotonicity are as follows.

Interval	$(-\infty, 0)$	$(0, 1)$	$(1, 2)$	$(2, \infty)$
Sign of f'	$+$	$-$	$-$	$+$
Variation of f	$\nearrow$	$\searrow$	$\searrow$	$\nearrow$

The first test gives a local maximum $f(0) = 0$ and local minimum $f(2) = 4$. The second derivative is negative for $x < 1$ and positive for $x > 1$. Thus the left branch is concave and the right branch convex. There is no inflection point: the change occurs at an input outside the domain. Neither local extremum is global on the whole domain, since the function is unbounded above and below near its vertical asymptote.

10.6 Optimising a box on its physical domain

Cut equal corner squares from a 20 cm by 12 cm rectangle and fold the remaining flaps to make an open box. Assume negligible material thickness. Let x be the cut length in centimetres. Figure 10.2 shows the cuts and fold lines.

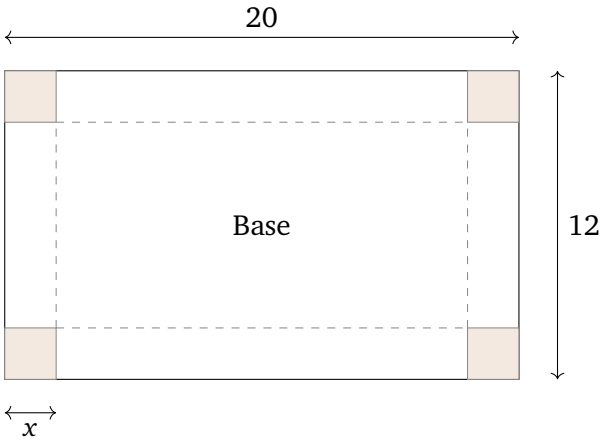


Figure 10.2: Four corner squares of side x are removed. Folding the dashed lines gives height x and base dimensions $20 - 2x$ and $12 - 2x$.

The volume in cubic centimetres is

$$V(x) = x(20 - 2x)(12 - 2x), \qquad 0 < x < 6.$$

The restrictions make all three lengths positive. For the maximum search extend this polynomial continuously to $[0, 6]$. Both endpoint values are zero, representing degenerate boxes. The extreme value theorem guarantees attainment on this closed interval. Its derivative is

$$V'(x) = 4(3x^2 - 32x + 60), \qquad r = \frac{16 - 2\sqrt{19}}{3}, \quad R = \frac{16 + 2\sqrt{19}}{3}.$$

The two displayed numbers are its roots. They satisfy $0 < r < 6 < R$. For example, $4 < \sqrt{19} < 5$ gives $2 < r < 8/3$ and $R > 8$. The quadratic is positive for $0 < x < r$ and negative for $r < x < 6$. Thus the unique maximum occurs at the physical cut length r. Substitution gives

$$V(r) = \frac{1792 + 1216\sqrt{19}}{27}\ \text{cm}^3.$$

The endpoint comparison and the physical restriction exclude both the larger algebraic root and a collapsed box.

Summary 10.

- Theorem 10.1: Cauchy's mean value identity compares two function changes. Its ratio form requires a nonvanishing denominator derivative.
- Theorem 10.2: the one-sided zero-over-zero rule is proved. The infinity-over-infinity form is cited. Each requires its own limiting form and an existing derivative-ratio limit.
- Propositions 10.3 and 10.4: derivative sign changes test local extrema. A nonzero second derivative at a stationary point also decides its type. A zero second derivative alone does not decide it.
- Theorem 10.6: a differentiable function is convex on an open interval exactly when its derivative is nondecreasing. For twice differentiable functions, this is equivalent to a nonnegative second derivative.
- A graph investigation keeps domain exclusions, asymptotes, derivative signs and convexity distinct. Optimisation also checks physical restrictions and boundary values before choosing among critical numbers.

Lecture 11 uses Cauchy's theorem to prove Taylor's remainder formula and quantify the errors in polynomial approximations. Sheet 10 practises limit rules, derivative tests and optimisation.

References

V. A. Zorich, *Mathematical Analysis I*, 2nd ed., Springer, 2015, §5.4. R. G. Bartle and D. R. Sherbert, *Introduction to Real Analysis*, 4th ed., Wiley, 2011, Chapter 6.